

phase and lower angular frequency than that in water.

17 B

$$s = r \theta$$

$$\theta = \frac{s}{r} = \frac{5.00}{500 \times 10^3} = 1.00 \times 10^{-5} \text{ rad}$$

$$\text{Min angle of resolution } \theta_m = \frac{\lambda}{b} = \frac{500 \times 10^{-9}}{10.0 \times 10^{-2}} = 5.00 \times 10^{-6} \text{ rad}$$

Images seen are resolved since angular separation between two sources is larger than the min angle of resolution.

18 B

Using Malus' law,